

S1. Supplementary Materials for TradingAgents

S1.1. Baseline Models

We compare our TradingAgents framework against several baselines:

- **Buy and Hold:** Investing equal amounts in all selected stocks and holding them throughout the simulation period.
- **MACD (Moving Average Convergence Divergence):** A trend-following momentum strategy that generates buy and sell signals based on the crossover points between the MACD line and signal line.
- **KDJ and RSI (Relative Strength Index):** A momentum strategy combining KDJ (stochastic oscillator) and RSI (relative strength index) indicators to identify overbought and oversold conditions for trading signals.
- **ZMR (Zero Mean Reversion):** A mean reversion trading strategy that generates signals based on price deviations from and subsequent reversions to a zero reference line.
- **SMA (Simple Moving Average):** A trend-following strategy that generates trading signals based on crossovers between short-term and long-term moving averages.

S1.2. Evaluation Metrics

To thoroughly evaluate the performance of our TradingAgents framework, we use widely recognized metrics to assess the risk management, profitability, and safety of the TradingAgents strategy in comparison to baseline approaches. Here we describe these metrics:

S1.2.1. Cumulative Return (CR)

The cumulative return measures the total return generated over the simulation period. It is calculated as:

$$CR = \left(\frac{V_{\text{end}} - V_{\text{start}}}{V_{\text{start}}} \right) \times 100\% \quad (\text{S1})$$

where V_{end} is the portfolio value at the end of the simulation, and V_{start} is the initial portfolio value.

S1.2.2. Annualized Return (AR)

The annualized return normalizes the cumulative return over the number of years:

$$AR = \left(\left(\frac{V_{\text{end}}}{V_{\text{start}}} \right)^{\frac{1}{N}} - 1 \right) \times 100\% \quad (\text{S2})$$

where N is the number of years in the simulation.

S1.2.3. Sharpe Ratio (SR)

The Sharpe ratio measures risk-adjusted return by comparing a portfolio's excess return over the risk-free rate to its volatility:

$$SR = \frac{\bar{R} - R_f}{\sigma} \quad (S3)$$

where $\bar{R}$ is the average portfolio return, R_f is the risk-free rate (e.g., yield of 3-month Treasury bills), and σ is the standard deviation of the portfolio returns.

S1.2.4. Maximum Drawdown (MDD)

Maximum drawdown measures the largest peak-to-trough decline in the portfolio value:

$$MDD = \max_{t \in [0, T]} \left(\frac{\text{Peak}_t - \text{Trough}_t}{\text{Peak}_t} \right) \times 100\% \quad (S4)$$

S1.3. Cumulative Returns (CR) and Transaction History for AMZN and GOOGL

We present additional figures for \$AMZN and \$GOOGL stocks to complement the AAPL data discussed in the main body of this paper. These supplementary visuals provide a broader perspective on the performance of our trading framework across multiple stocks, highlighting the consistency and robustness of TradingAgents's results.

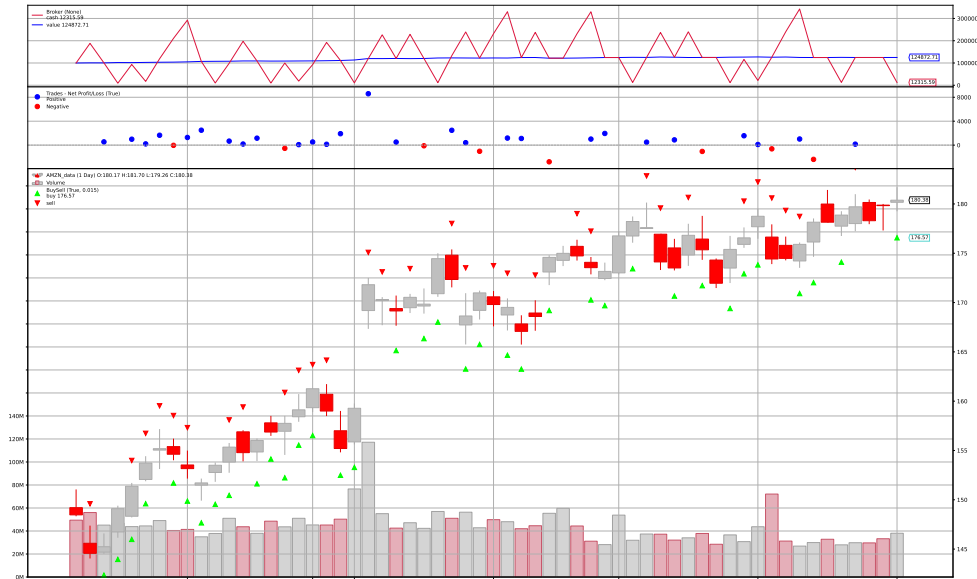


Fig. S1: **TradingAgents**: Transaction History for AMZN. Detailed transaction history with Green / Red arrows for Long / Short positions.

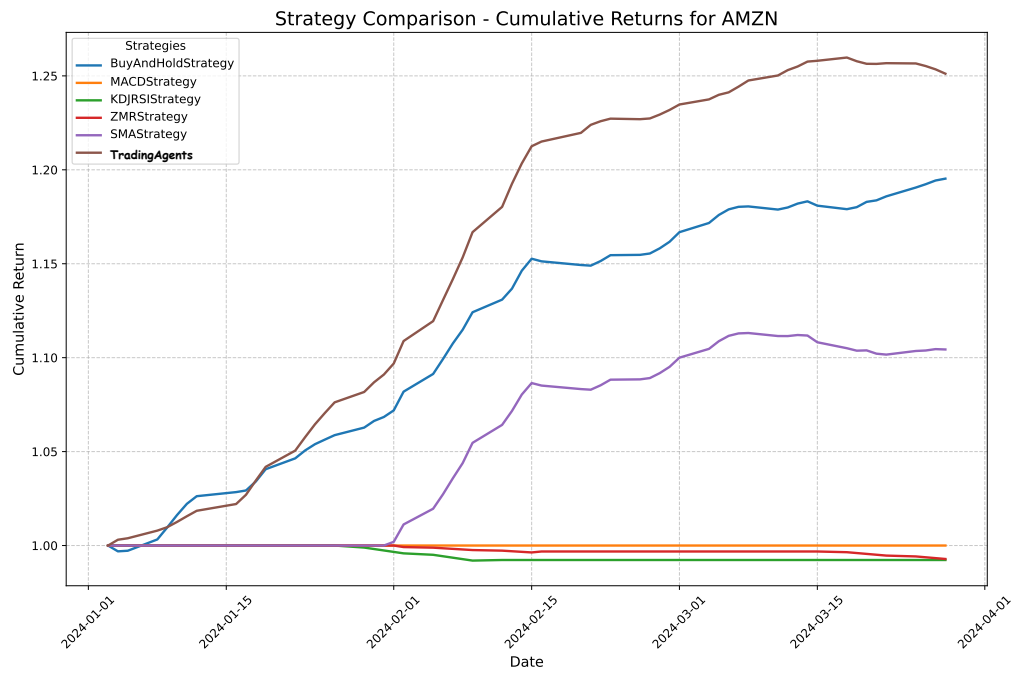


Fig. S2: **TradingAgents**: Cumulative Returns for AMZN. Performance comparison showing cumulative returns.

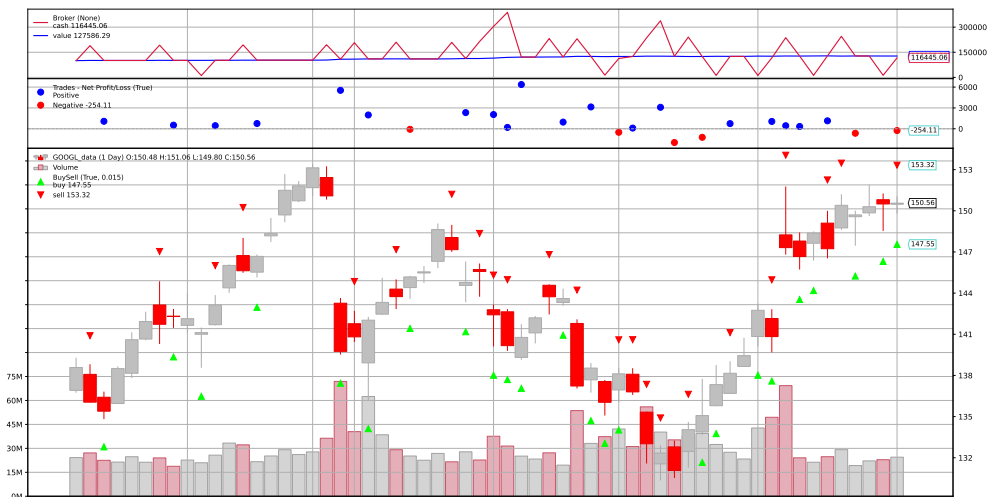


Fig. S3: **TradingAgents**: Transaction History for GOOGL. Detailed transaction history with Green / Red arrows for Long / Short positions.